

Parameter	Explicit choice	Scaling in r
M	2^{27}	Fixed
γ	$1 - 10^{-6}$	Fixed
C_1	2×10^8	Fixed
$C_{2,r}$	$10001\sqrt{2}\sqrt{(M+9)^r - M^r}$	$\Theta((M+9)^{r/2})$
L_r	$\left\lceil \frac{\log(2P_r/E)}{2\log\theta} \right\rceil$	$\Theta(rM^{2r})$
Q_r	2^{q_r} , with q_r in (S42)	$\exp(\Theta(r^2M^{2r}))$
Input dimension n_r	$k^2Q_r^{2r}$	$\exp(\Theta(r^3M^{2r}))$
Output dimension k	M^r	$\exp(\Theta(r))$

Table II: Parameters of the final channel $\hat{\Lambda}_r$. The basic channel Φ_r has input dimension Q_r^{2r} . The quantities E, θ, P_r are defined below.

Proof of the main theorem

We use the parameters in Table II. Thus $M = 2^{27}$, $\gamma = 1 - 10^{-6}$, and $C_1 = 2 \times 10^8$. Throughout this proof, $r \geq 1$ is an integer. Put

$$k = M^r, \quad A_r = (M+9)^r - M^r, \quad E = 10^{-6}, \quad \theta = \frac{10001}{10000}, \quad P_r = \left(2 \left\lceil \sqrt{C_1} k \right\rceil\right)^{k(k-1)}.$$

Choose

$$C_{2,r} = 10001\sqrt{2A_r}, \quad L_r = \left\lceil \frac{\log(2P_r/E)}{2\log\theta} \right\rceil, \quad (S42)$$

$$Q_r = 2^{q_r}, \quad q_r = 2 + \lceil L_r [4\log_2(4M-1) + 2\log_2 k] \rceil.$$

These choices satisfy $C_{2,r} > \sqrt{C_1 A_r}$, $L_r \geq 1$, and give $P_r \theta^{-2L_r} \leq E/2$ and $Q_r > B_{L_r} k^{2L_r}$. Since $A_r \geq 1$, Proposition 3 yields

$$\frac{\text{Tr } F}{Q_r^{2r}} \leq P_r \theta^{-2L_r} \left[1 + \frac{B_{L_r}}{Q_r} \left(\frac{k(k-1)}{A_r} \right)^{L_r} \right] < E. \quad (S43)$$

Let Φ_r be the resulting basic channel and write

$$a_r = 1 + \frac{\gamma^2}{k} \left(\frac{C_{2,r}}{\sqrt{C_1} - \sqrt{2}} \right)^2. \quad (S44)$$

The matrices U_I are real, and the grid $\mathcal{W}_k$ is closed under entrywise conjugation. Hence F and $H = \sqrt{\gamma}(I+F)^{-1/2}$ are real, so $\overline{\Phi_r} = \Phi_r$. Proposition 2, Lemma 3, and Corollary 1 give

$$k \left\| \Phi_r(\sigma) - \frac{I_k}{k} \right\|_2^2 \leq a_r - 1, \quad H((\Phi_r \otimes \Phi_r)(\omega_{Q_r^{2r}})) \leq 2\log k - rd_M \left(\frac{\gamma^2}{(1+E)^2} \right), \quad (S45)$$

where $\omega_{Q_r^{2r}}$ is the maximally entangled input state defined above.

We next make every one-copy output close to the maximally mixed state using a short average of Pauli conjugations. Set

$$m = 27r, \quad t_r = \lceil \log_2(54r^2 \sqrt{a_r}) \rceil, \quad h_r = 2^{t_r}, \quad \ell_r = h_r^2.$$

Represent $\mathbb{F}_{h_r}$ using the lexicographically first monic irreducible binary polynomial of degree t_r , and let $\tau : \mathbb{F}_{h_r} \rightarrow \mathbb{F}_2$ be its field trace. For $x, y \in \mathbb{F}_{h_r}$, define

$$P_{x,y} = \bigotimes_{j=1}^m X^{\tau(yx^{j-1})} Z^{\tau(yx^{m+j-1})}, \quad \mathcal{R}_r(Y) = \frac{1}{\ell_r} \sum_{x,y} P_{x,y} Y P_{x,y}^\dagger.$$

Here X, Z are the qubit Pauli matrices and $x^0 = 1$. For every nonidentity element P of the Pauli basis, conjugation contributes a sign $(-1)^{\tau(y p(x))}$ for a nonzero binary polynomial p of degree at most $2m - 1$. Averaging first over y gives

$$\mathcal{R}_r(P) = \frac{\#\{x : p(x) = 0\}}{h_r} P, \quad \frac{\#\{x : p(x) = 0\}}{h_r} \leq \frac{2m - 1}{h_r} < \frac{1}{r\sqrt{a_r}}.$$

Orthogonality of the Pauli basis and the definition of t_r imply

$$\|\mathcal{R}_r(Y)\|_2 \leq \frac{\|Y\|_2}{r\sqrt{a_r}} \quad (\text{Tr } Y = 0), \quad \ell_r < 108^2 r^4 a_r. \quad (\text{S46})$$

Consequently $\Lambda_r = \mathcal{R}_r \circ \Phi_r$ satisfies

$$k \left\| \Lambda_r(\sigma) - \frac{I_k}{k} \right\|_2^2 \leq \frac{a_r - 1}{r^2 a_r} \leq r^{-2}.$$

The inequality $H(\rho) \geq -\log \text{Tr}(\rho^2)$ now gives

$$H_{\min}(\Lambda_r) \geq \log k - \log(1 + r^{-2}). \quad (\text{S47})$$

On the other hand, $(\mathcal{R}_r \otimes \mathcal{R}_r)$ averages ℓ_r^2 unitary conjugations. Applying the entropy-of-mixture inequality to the Bell output in (S45) gives

$$H((\Lambda_r \otimes \Lambda_r)(\omega_{Q_r^{2r}})) \leq 2 \log k - r d_M \left(\frac{\gamma^2}{(1 + E)^2} \right) + 2 \log \ell_r. \quad (\text{S48})$$

We obtain the final channel by appending a classical label. For $u, v \in \{0, 1\}^m$, write $W_{uv} = \bigotimes_{j=1}^m X^{u_j} Z^{v_j}$, and define

$$\hat{\Lambda}_r(\rho) = \sum_{u,v} W_{uv} \Lambda_r(\langle u, v | \rho | u, v \rangle) W_{uv}^\dagger. \quad (\text{S49})$$

Its input and output dimensions are $k^2 Q_r^{2r}$ and k . Every output is a mixture of unitary conjugates of outputs of Λ_r . Entropy concavity therefore gives

$$\chi_{\max}(\hat{\Lambda}_r) \leq \log k - H_{\min}(\Lambda_r).$$

For two copies, use the Bell input and choose both labels independently and uniformly. The average output is I_{k^2}/k^2 , while all states in this ensemble have the same entropy. Hence

$$\chi_{\max}(\hat{\Lambda}_r^{\otimes 2}) \geq 2 \log k - H((\Lambda_r \otimes \Lambda_r)(\omega_{Q_r^{2r}})) \geq r d_M \left(\frac{\gamma^2}{(1 + E)^2} \right) - 2 \log \ell_r.$$

It remains to check that the two-copy gain exceeds the cost of this short average. Put $\beta = (10001\gamma/9999)^2$. Then $1 < \beta < 1.001$, and the binomial theorem gives

$$a_r = 1 + \beta \left[\left(1 + \frac{9}{M} \right)^r - 1 \right] \leq \left(1 + \frac{9\beta}{M} \right)^r.$$

For $s = \gamma^2/(1+E)^2$, the elementary estimate

$$d_M(s) \geq \frac{s \log M + s \log s + (1-s) \log(1-s)}{M}$$

and $-s \log s \leq 1-s$ imply

$$\begin{aligned} d_M(s) - 2 \log \left(1 + \frac{9\beta}{M} \right) &\geq \frac{\log M - (1-s)[\log M + 1 - \log(1-s)] - 18\beta}{M} \\ &> \frac{0.692868}{M} > 5 \times 10^{-9}. \end{aligned}$$

Here we used only $18.711 < \log M < 19$, $1-s < 4 \times 10^{-6}$, $\log(1-s) > -13$, and $\beta < 1.001$. The first estimate follows by writing $1 + (M-1)s = M(s + (1-s)/M)$ and using $(x+y) \log(x+y) \geq x \log x + y \log y$ for $x, y > 0$. Combining the preceding bounds proves

$$\chi_{\max}(\widehat{\Lambda}_r) \leq \log(1+r^{-2}), \quad \chi_{\max}(\widehat{\Lambda}_r^{\otimes 2}) > 5 \times 10^{-9}r - 8 \log r - 4 \log 108. \quad (\text{S50})$$

The sharper asymptotic coefficient follows from $a_r = 1 + \beta[(1+9/M)^r - 1] \leq \beta(1+9/M)^r$:

$$\chi_{\max}(\widehat{\Lambda}_r^{\otimes 2}) \geq r \left[d_M(s) - 2 \log \left(1 + \frac{9}{M} \right) \right] - 8 \log r - 4 \log 108 - 2 \log \beta. \quad (\text{S51})$$

The corresponding upper bound holds for the entropy of the actual Bell output of $\Lambda_r^{\otimes 2}$, by (S48). Thus the remainder in the stated asymptotic estimate is explicitly controlled by $8 \log r + 4 \log 108 + 2 \log \beta$.

Given $N \geq 1$, take $J = \lceil N \rceil$ and $r_N = 10^{11}J$, and define $\Phi_N = \widehat{\Lambda}_{r_N}$. Since $\log(1+x) < x$ for $x > 0$, the first bound in (S50) is less than $r_N^{-2} < 1/N$. Using $\log(10^{11}) < 26$, $4 \log 108 < 20$, and $\log J \leq J-1$, the second bound is greater than

$$500J - 8(26 + \log J) - 20 \geq 492J - 220 > N.$$

Also $C(\Phi_N) \geq \frac{1}{2} \chi_{\max}(\Phi_N^{\otimes 2})$, so $C(\Phi_N) - \chi_{\max}(\Phi_N) > N/2 - 1/N$. Finally, $\log P_r = \Theta(rM^{2r})$ gives the scalings in Table II. Since $r_N = \Theta(N)$, the output dimension is $\exp(\Theta(N))$ and the input dimension is $\exp(\exp(\Theta(N)))$, as claimed.